



↻ WHILE Loop – 3 Complexity Levels in T-SQL

✓ LEVEL 1 – Basic WHILE Loop

🎯 Goal: Print numbers from 1 to 5

```
DECLARE @Counter INT = 1

WHILE @Counter <= 5
BEGIN
    PRINT 'Counter value: ' + CAST(@Counter AS NVARCHAR)
    SET @Counter = @Counter + 1
END
```

📖 *Introductory logic to understand WHILE loop syntax and incrementation.*

✓ LEVEL 2 – Intermediate WHILE Loop

🎯 **Goal: Print the first 5 email addresses from Person.EmailAddress**

```
DECLARE @Counter INT = 1
DECLARE @Max INT = 5
DECLARE @Email NVARCHAR(50)

DECLARE email_cursor CURSOR FOR
    SELECT TOP (@Max) EmailAddress
    FROM Person.EmailAddress
    ORDER BY BusinessEntityID

OPEN email_cursor
FETCH NEXT FROM email_cursor INTO @Email

WHILE @Counter <= @Max AND @@FETCH_STATUS = 0
BEGIN
    PRINT 'Email ' + CAST(@Counter AS NVARCHAR) + ': ' + @Email
    SET @Counter = @Counter + 1
    FETCH NEXT FROM email_cursor INTO @Email
END

CLOSE email_cursor
DEALLOCATE email_cursor
```

📖 Combines `WHILE` with a cursor. Emphasizes cursor iteration control using `@@FETCH_STATUS`.

✓ LEVEL 3 – Advanced WHILE Loop

🎯 **Goal: Calculate total sales for the first 10 orders, one by one, and stop if total exceeds \$10,000**

```
DECLARE @OrderID INT
DECLARE @Total MONEY = 0
DECLARE @RunningTotal MONEY = 0
DECLARE @Counter INT = 1
DECLARE @Max INT = 10

DECLARE order_cursor CURSOR FOR
    SELECT TOP (@Max) SalesOrderID, TotalDue
    FROM Sales.SalesOrderHeader
    ORDER BY SalesOrderID

OPEN order_cursor
FETCH NEXT FROM order_cursor INTO @OrderID, @Total

WHILE @@FETCH_STATUS = 0
BEGIN
    SET @RunningTotal += @Total
    PRINT 'Order ' + CAST(@OrderID AS NVARCHAR) + ' Total: $' + CAST(@Total AS NVARCHAR)
    PRINT 'Running Total: $' + CAST(@RunningTotal AS NVARCHAR)

    IF @RunningTotal > 10000
    BEGIN
        PRINT 'Stopping loop: Total sales exceeded $10,000'
        BREAK
    END

    SET @Counter += 1
    FETCH NEXT FROM order_cursor INTO @OrderID, @Total
END

CLOSE order_cursor
DEALLOCATE order_cursor
```

 Demonstrates WHILE with `BREAK` , accumulation logic, and business rules.

Summary

Level	Focus	Concepts Used
1	Counter & loop	<code>WHILE</code> , <code>PRINT</code> , <code>SET</code>
2	Data row iteration	<code>WHILE</code> , <code>CURSOR</code> , <code>@@FETCH_STATUS</code>
3	Business logic with break condition	<code>WHILE</code> , <code>CURSOR</code> , <code>BREAK</code> , totals
